

Mission Messages and Binary

Name: _____

Date: _____

1. Read the four-flash language

EASY

In the mission codebook, red means 0 and green means 1. Use the reference codebook on the final page.

1. Decode each transmission: 0001 = _____ 1000 = _____
2. Write the code for each report: Salt detected = _____ Return to shelter = _____
3. Circle every pair that differs in exactly one position: **1010/1011** **0000/1111** **0100/0101**
4. A probe sends 1100. One flash flips. Write all four possible received strings.

2. How many messages fit?

EASY

Complete the doubling table.

Number of flashes n	1	2	3	4	5
Possible strings 2^n	2				

Why are three flashes insufficient for sixteen reports?

Why are four flashes exactly enough?

3. Distance measures danger

EASY

The Hamming distance is the number of positions where two strings differ.

- $d(1010, 1011) = \underline{\hspace{2cm}}$ $d(0000, 1110) = \underline{\hspace{2cm}}$ $d(0110, 1001) = \underline{\hspace{2cm}}$
- Which pair above is most easily confused by one flipped bit? Explain.

- Mark the differing positions: 1 0 0 1 and 1 1 1 0. Distance = _____

6. Build and defend a code

CHALLENGE

Choose three valid five-bit messages. Keep every pair at distance at least 3.

Message	Five-bit code	Why it is far enough from the others
A		
B		
C		

For each message, write one string produced by flipping a single bit. Check that none of your examples could be interpreted as a one-bit error from two different messages.

Detect Repeat or Repair

Name: _____

Date: _____

1. The parity watchdog

EASY

Append one parity bit so the total number of 1s is even.

Four data bits	Parity bit	Five transmitted bits
1010		
1110		
0001		
1101		

a) The receiver gets 10 101. Is its total parity even or odd? _____

b) What can the watchdog conclude?

c) Can it identify which position changed? Explain.

3. Repeat the whole message

EASY

The original four-bit report is sent three times.

Copy	Bit 1	Bit 2	Bit 3	Bit 4
1 - original message	1	0	1	0
2	1	0	1	1
3	0	0	1	0
Column majority				

Recovered message: _____. Use the codebook on the final page.

Decoded report: _____

How many transmitted bits were used? _____ How many were extra copies? _____

4. When majority voting fails

CHALLENGE

Create three copies of one four-bit message so that two errors in the same column make majority voting return the wrong bit. Show the original and all three received copies.

Original: _____ Copy 1: _____ Copy 2: _____ Copy 3: _____

State precisely what triple repetition guarantees and what it does not guarantee.

Detect Repeat or Repair

6. Communication budget

CHALLENGE

A probe can transmit at most 70 bits during a short communication window.

a) How many four-bit reports fit with no protection?

b) How many fit if every report is repeated three times?

c) How many fit using Hamming(7,4)?

d) Which method would you choose if single-bit correction is required? Defend the tradeoff.

7. Burst-error thought experiment

CHALLENGE

A radiation burst flips three adjacent transmitted bits. Compare two layouts:

- Layout A sends each three-copy message consecutively.
- Layout B interleaves bits from many different messages.

Which layout is more likely to spread the damage across different protected blocks? Why can spreading errors help?

Hamming(7,4) Inspect and Repair

Name: _____

Date: _____

1. Give every position an address

EASY

Write the inspector answers in the order Inspector 4, Inspector 2, Inspector 1.

Position	Binary address	Inspector 1?	Inspector 2?	Inspector 4?
1	001	yes	no	no
2	010	no	yes	no
3	011			
4	100			
5	101			
6	110			
7	111			

Why does 000 mean “no error”?

Why are positions 1, 2, and 4 natural places for the check bits?

2. Place data and checks

EASY

Fill the roles in order. Use P for a parity/check bit and D for a data bit.

1	2	3	4	5	6	7

Complete the watcher sets:

Inspector 1 watches _____; Inspector 2 watches _____; Inspector 4 watches _____.

3. Encode report 1010

EASY

Put data 1010 into positions 3, 5, 6, and 7. Choose P1, P2, and P4 so every inspector sees an even number of 1s.

P1	P2	D1	P4	D2	D3	D4
		1		0	1	0

Inspector calculations: P1 = _____; P2 = _____; P4 = _____.

Seven-bit codeword: _____.

Hamming(7,4) Inspect and Repair

4. Locate one damaged bit

EASY

Received word: **1011000**. Test even parity.

Inspector	Positions checked	Number of 1s	Satisfied or complains?
1	1, 3, 5, 7		
2	2, 3, 6, 7		
4	4, 5, 6, 7		

Add the labels of complaining inspectors: _____. Error position: _____.

Flip that bit. Repaired codeword: _____. Extract positions 3, 5, 6, 7: _____.

Use the codebook on the final page. Report: _____.

Limits Noise and Research

Name: _____

Date: _____

1. Count each error neighborhood

EASY

For an n -bit protected message, the receiver may see the original or any one of n single-bit flips. Each correctable message therefore needs $n + 1$ exclusive received strings.

n	All strings 2^n	Slots per message $n + 1$	$16(n + 1)$ needed	Can 16 fit?
5	32	6		
6	64	7		
7	128	8		

Why can no clever rearrangement make five or six bits sufficient?

What is special about $16(7 + 1) = 2^7$?

3. Transmission cost

EASY

Hamming(7,4) converts every four data bits into seven transmitted bits.

a) Extra bits per four-bit message: _____

b) Transmitted bits for one 8-bit byte: _____

c) The complete text used in the slides has 3,359,405 bytes. Confirm that encoding it uses 47,031,670 transmitted bits. Show the multiplication.

d) Compare with repeating every bit three times. How many transmitted bits would that require?

Limits Noise and Research

5. Predict recovery with probability

CHALLENGE

For independent flip probability p , a seven-bit codeword has no errors with probability $(1 - p)^7$ and exactly one error with probability $7p(1 - p)^6$.

1. Write the probability that Hamming(7,4) is within its guarantee:

P(0 or 1) =

2. Evaluate it for $p = 0.03$.

3. A byte uses two codewords. Estimate the probability that both are within the guarantee by squaring your answer.

4. Repeat for $p = 0.10$. What changes most sharply?

7. Design a research test

CHALLENGE

Choose one question.

- How does exact text recovery change as the flip rate moves from 0 to 20 percent?
- Can a different seven-bit code protect more than sixteen four-bit reports?
- How should a code change when two errors must be corrected?
- How does interleaving change damage from a fixed-length burst?

Claim or prediction:

Variables and test procedure:

Evidence to record:

What result would change your mind?

Original NILA Mission Codebook

The sixteen four-bit reports used throughout the worksheets.

Code	Mission report	Code	Mission report
0000	Systems normal	1000	Liquid water detected
0001	Battery low	1001	Organic molecules detected
0010	Camera ready	1010	Possible biological pattern
0011	Camera damaged	1011	Mission emergency
0100	Drill ready	1100	Radiation spike
0101	Drill jammed	1101	Surface crack opening
0110	Ice sample collected	1110	Warm plume detected
0111	Salt detected	1111	Return to shelter

Hamming(7,4) reference

Check positions: 1, 2, 4 Data positions: 3, 5, 6, 7

Inspector 1 watches 1, 3, 5, 7. Inspector 2 watches 2, 3, 6, 7. Inspector 4 watches 4, 5, 6, 7.

Counting bound

If M length- n codewords must correct every one-bit error, their radius-one neighborhoods cannot overlap, so $M(n + 1) \leq 2^n$. Equality for $M = 16$ and $n = 7$ makes Hamming(7,4) a perfect code.